

BAAZ & UDAY JKBOSE

2026 CLASS 11TH

STRUCTURE OF ATOM

LECTURE-1

CHEMISTRY

VISHAL SIR

JKBose Wallah



Topics To Covered

- 1** **Discovery of Electron**
- 2** **Discovery of Proton**

REVISION

STOICHIOMETRY

CONCENTRATION

LIMITING REAGENT.
OR
Reactant.

QUESTION



What volume of oxygen at S.T.P. is required to effect complete combustion of 200 ^{cm³} of acetylene and what would be the volume of carbon dioxide formed?
($\text{HC}\equiv\text{CH}$.)

$$1\text{L} = 1000\text{cm}^3$$

$$1\text{cm}^3 = \frac{1}{1000}\text{L or } 10^{-3}\text{L}$$

$$200\text{cm}^3 = \frac{200}{1000}\text{L} = 0.2\text{L}$$

Sol. Balanced Chemical Equation



Molar Volume	22.4L	2.5 × 22.4L	2 × 22.4L	22.4L
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Given Volume	0.2L	?
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• 22.4L of C_2H_2 require = 2.5 × 22.4L of oxygen.

1L of " " = $\frac{2.5 \times 22.4\cancel{\text{L}}}{22.4\cancel{\text{L}}}$ of oxygen

0.2L of " " = 2.5 × 0.2L of oxygen = 0.5L of oxygen.

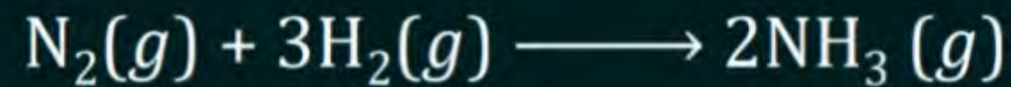
• Volume of CO_2 .

22.4L of C_2H_2 produce = 44.8L of CO_2 .

1L of " " = $\frac{44.8}{22.4}$ L of CO_2

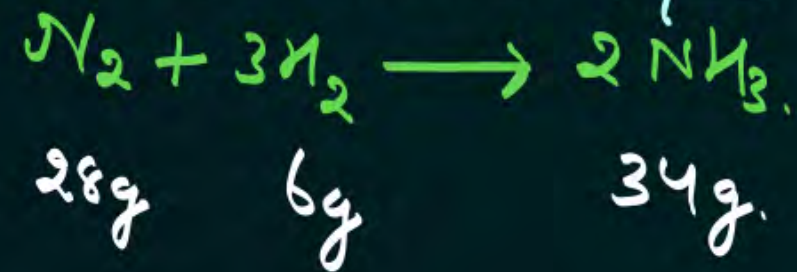
0.2L of C_2H_2 produce = 2 × 0.2L of CO_2
= 0.4L of CO_2 .

Dinitrogen and dihydrogen react with each other to produce ammonia according to the following chemical equation,



- Calculate the mass of ammonia produced if $2.00 \times 10^3 g$ dinitrogen reacts with $1.00 \times 10^3 g$ of dihydrogen.
- Will any of the two reactants remain unreacted? (excess Reagent).
- If yes, which one and what would be its mass?

Sol: → Balanced chemical equation.



Molar
mass

28g 6g 34g

Given
mass

$2 \times 10^3 g$ $1 \times 10^3 g$
 of
 1000g

28g of N_2 require = 6g of H_2 .

1g " " " = $\frac{6}{28} g$ of H_2 .

$2 \times 10^3 g$ " " " = $\frac{6}{28} \times 2 \times 10^3 g$ of H_2 .

= 428.57g of H_2 .

∴

Hence, N_2 is limiting is Reagent & H_2 is excess Reagent.

Amount of excess of
Reagent $\Rightarrow$ Given amount - Require amount.

$$= 1000 \text{ g} - 428.57 \text{ g}$$

$$= 571.43 \text{ g}.$$

CONCENTRATION TERMS.

Terms used to express the amount of solute and solvent present in a solution.
(Component)

$$1) \text{ Mass percentage} = \frac{\text{Mass of that component}}{\text{Total mass of solution}} \times 100.$$

$$2) \text{ Mole fraction } (X) = \frac{\text{No. of moles of that component in solution}}{\text{Total no. of moles of solution}}$$

Let A is solute & B is solvent

$$X_A = \frac{n_A}{n_A + n_B}$$

$$X_B = \frac{n_B}{n_A + n_B}$$

* Total molefraction of solution i.e. $X_A + X_B = 1$

$$3) \text{ Molarity } (M) = \frac{\text{No. of moles of solute}}{\text{Total volume of solution (l)}}.$$

$$4) \text{ Molality } (m) = \frac{\text{No. of moles of solute}}{\text{Mass of solvent (kg)}}.$$

UNIT-II: STRUCTURE OF ATOM

9 MARKS

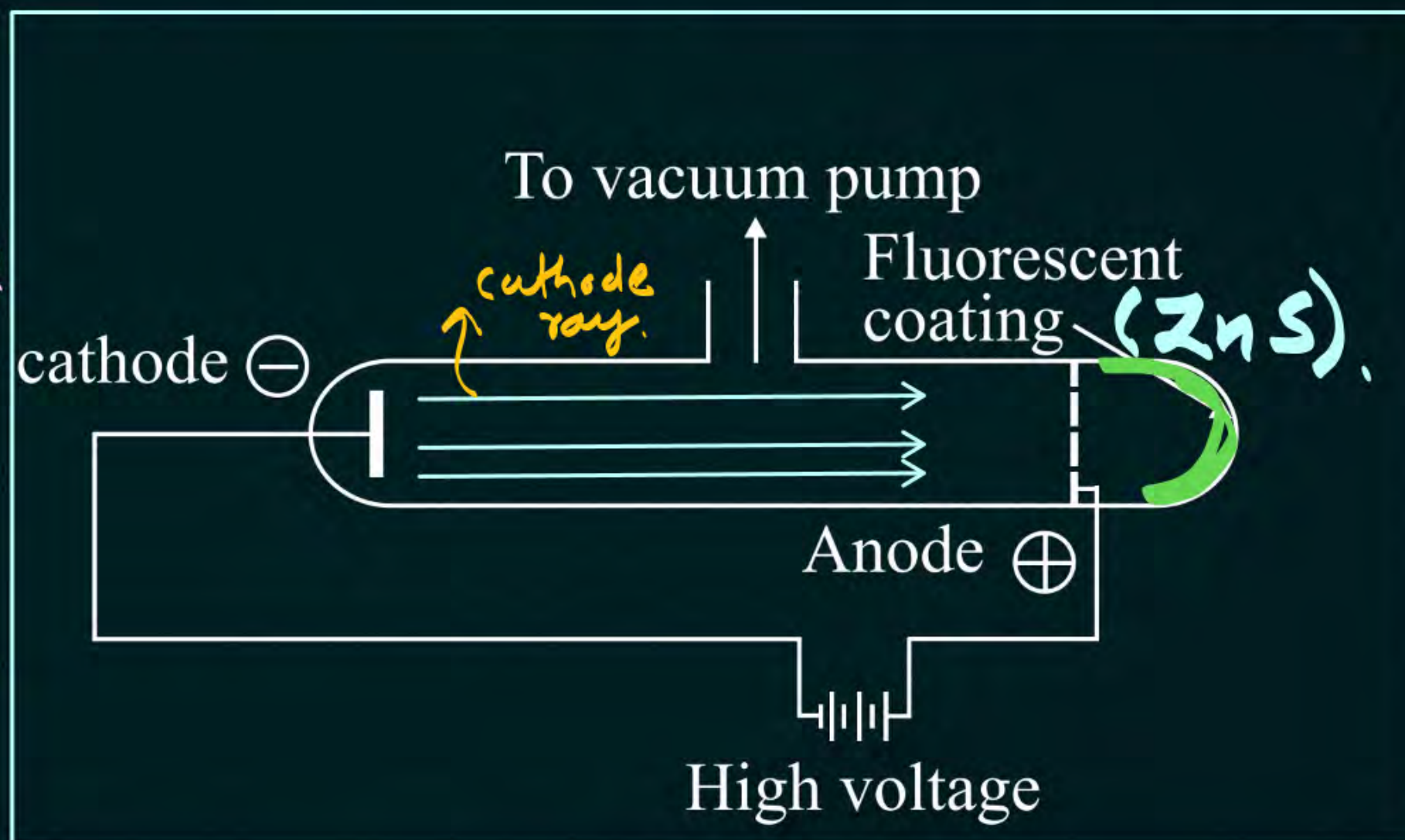
- Discovery of electron, proton and neutron, atomic number, isotopes and isobars. Thompson's model and its limitations, Rutherford's model and its limitations. Bohr's model & its limitations, concept of shells and sub-shells. Dual nature of matter and light, de-Broglie's relationship. Heisenberg's uncertainty principle, concept of orbitals, quantum numbers, shapes of s, p and d-orbitals. Rules for filling electrons in orbitals- Aufbau's principle, Pauli's exclusion principle and Hund's rule. Electronic configuration of atoms, stability of half-filled and completely filled orbitals.

Discovery of Electron



- Discovery of electron was done by J. J. Thompson in late 19th century through discharge tube experiment.

• Discharge tube or cathode ray tube is a glass tube having two metal rods (electrode) at the corner of tube and gas is placed inside it at low pressure & high voltage. The electrodes are connected with the battery and when voltage was applied a stream of particle called cathode ray was observed flowing from negative rod (cathode) to positive electrode (anode).



To confirm the flow of cathode rays - perforated anode was used and area behind the anode was coated with a fluorescent material (ZnS). When the rays pass through the holes & strike the ZnS they produce bright spot.

• → PROPERTIES OF CATHODE RAYS

- 1) Cathode rays travel in straight line.
- 2) Cathode rays are made up of material particle.
- 3) Cathode rays produce heating effect.
- 4) " " Cause ionisation of the gas through which they pass.

5) When the external electric field is applied on cathode rays, they deflect towards the positive plate of electric field. This shows they carry negative charge.

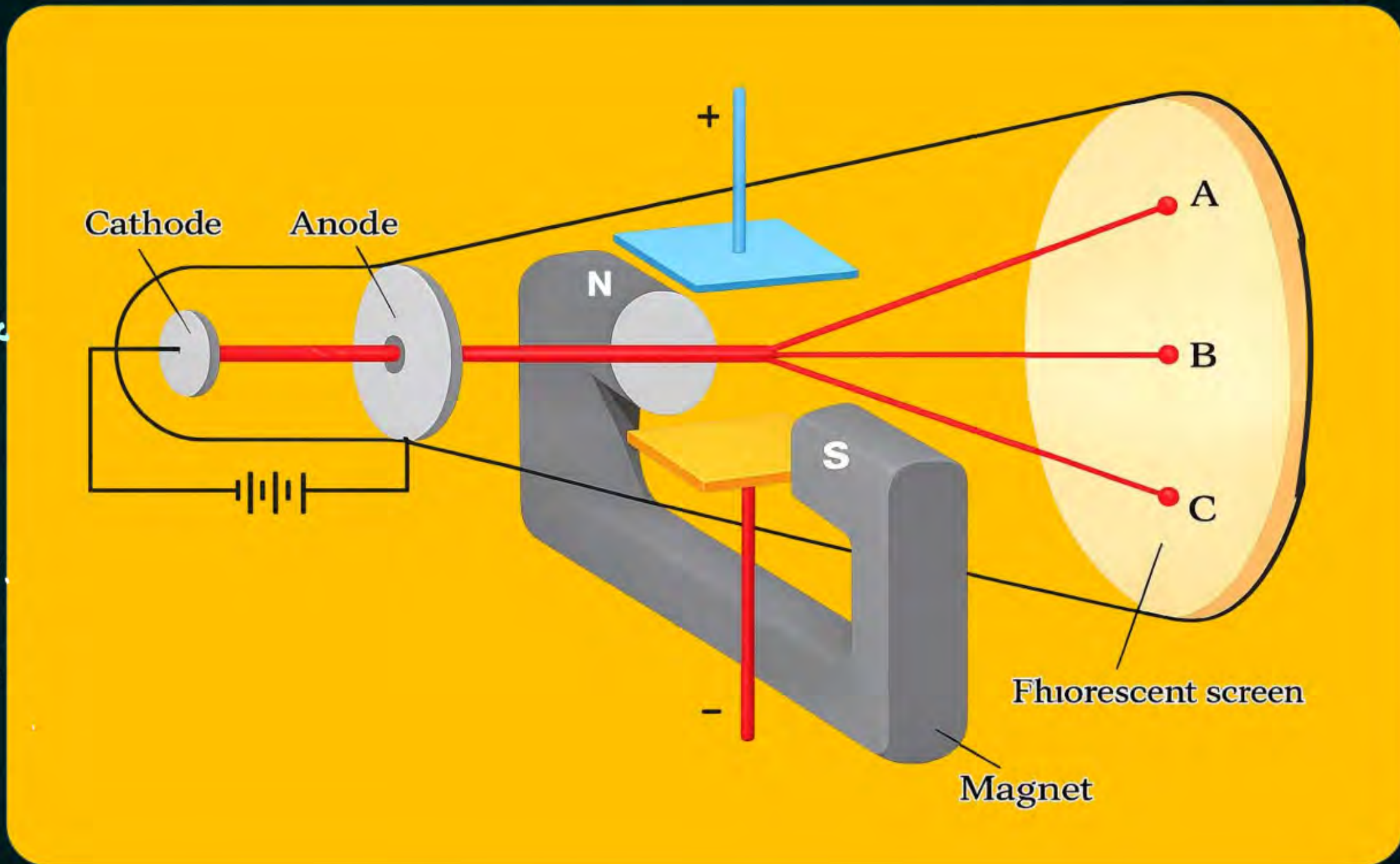
6) The characteristics of the cathode rays does not depend upon the nature of electrode & type of gas placed in discharge tube.

CHARGE TO MASS RATIO OF ELECTRON OR SPECIFIC CHARGE OF ELECTRON. (e/m_e)

• It was calculated by J. J. Thompson by using cathode ray tube with perpendicular magnetic and electric field.

• Key observations are.

- With only electric field $\vec{E}$'s deviate to point A.
- With only magnetic field $\vec{B}$, deviate towards point C.
- By balancing both fields $\vec{E}$ move straight and strike point B.



Deflection of Charged Particles $d \propto \frac{qE}{m}$

- d) Greater charge causes more deflection.
- e) Lighter particles deflected more.
- f) Stronger electric and magnetic field increases deflection.

By doing accurate measurements Thompson calculated charge to mass ratio

$$\frac{e}{m_e} = 1.758 \times 10^{11} \text{ C Kg}^{-1}$$

OR
1.76

CHARGE OF ELECTRON

R. A. Milliken determined the charge of electron with the help of **OIL DROP EXPERIMENT**. He found the charge on the electron to be $1.6 \times 10^{-19} \text{ C}$.

Mass of e^- was also calculated as.

$$\frac{e}{m_e} = 1.75 \times 10^{11} \text{ C Kg}^{-1}$$

$$\text{or } \frac{1.6 \times 10^{-19} \text{ C}}{1.75 \times 10^{11} \text{ C/Kg}} = m_e$$

$$\therefore m_e = 9.1 \times 10^{-31} \text{ kg}$$

$$\frac{e}{m_e} = 1.758 \times 10^{11} \text{ C kg}^{-1}$$

$$e = 1.6 \times 10^{-19} \text{ C}$$

$$m_e = 9.1 \times 10^{-31} \text{ kg.}$$

QUESTION



Which of the following scientists discovered the electron?

1 Ernest Rutherford

2 J.J. Thomson

3 Niels Bohr

4 James Chadwick

QUESTION

In the cathode ray discharge tube experiment, the cathode rays originate from:

- 1 Anode
- 2 Cathode
- 3 Fluorescent screen
- 4 Vacuum pump

QUESTION*or specific charge of e^-*

The charge-to-mass ratio (e/m) of an electron is approximately:

1

$$1.76 \times 10^{11} \text{ C/kg}$$

2

$$1.76 \times 10^8 \text{ C/kg}$$

3

$$1.76 \times 10^{12} \text{ C/kg}$$

4

$$1.76 \times 10^9 \text{ C/kg}$$

FIRST CHAPTER → TAG DA REVISION.

March, (3 March)
Revise }

REVISE KARKE. Aana.

Thank
You